Advanced Quantitative Research Methods

Conformal Prediction

Advanced module A3 — optional self-study

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Slides, lab notebooks and data:
github.com/ChrisW09/Advanced-Quantitative-Research-Methods

Where this module fits

The twelve **advanced modules** are optional self-study decks that sit outside the taught chapters. This one turns any predictive model from the course into a model with *honest uncertainty*.

You already know (chapter)	What module A3 adds
Ch. 3 — prediction intervals for linear regression	Intervals with *guaranteed* coverage, without Gaussian errors or a correct model
Ch. 5 — train / validation splits	A third role for held-out data: the *calibration set*
Ch. 4 — classifiers and their probabilities	Prediction *sets*: "{flu, covid}, with 90% guarantee"
Ch. 8 — gradient boosting	Boosted *quantile* regression as the engine of adaptive intervals (CQR)

Prerequisites: Chapters 3 and 5 are essential, Chapters 4 and 8 helpful. Short exercises every ∼20 min; extended exercises every ∼45 min; the module has a companion lab notebook.

Contents

1. Why Conformal
2. Split Conformal
3. Adaptive Intervals
4. Conformal Classification
5. Conformal vs Classical
6. Pitfalls and Practice
7. Python Lab
8. Summary

Optional and advanced material is collected in the **appendix** at the end of the deck.

Notation in this module

Symbol	Meaning
(x_i, y_i)	feature–response pair of observation i
$\mathcal{D}_{\text{tr}}$, $\mathcal{D}_{\text{cal}}$	proper training set and calibration set (disjoint)
n	number of calibration points, $n = \lvert\mathcal{D}_{\text{cal}}\rvert$
$\hat{f}$	prediction rule fitted on $\mathcal{D}_{\text{tr}}$ only
s_i	conformity score of calibration point i (e.g. $\lvert y_i - \hat{f}(x_i)\rvert$)
$s_{(k)}$	k-th smallest calibration score (order statistic)
α	target miscoverage level (we use $\alpha = 0.1$: 90% coverage)
$\hat{q}$	conformal quantile: $s_{(k)}$ with $k = \lceil (n+1)(1-\alpha) \rceil$
$C(x)$	prediction interval (regression) or prediction set (classification)
$\rho_\tau(u)$	pinball loss of quantile regression at level τ
$\hat{q}_{\text{lo}}, \hat{q}_{\text{hi}}$	fitted conditional quantile curves (levels $\alpha/2$, $1-\alpha/2$)
$\hat{p}_k(x)$	estimated probability of class k at x (softmax output)
K	number of classes

Sources and references

Primary textbook

James, G., Witten, D., Hastie, T., Tibshirani, R., & Taylor, J. (2023). *An Introduction to Statistical Learning, with Applications in Python.* Springer. (Chapters 3 and 5 supply the prerequisites.)

Further reading on conformal prediction

Vovk, V., Gammerman, A., & Shafer, G. (2005). *Algorithmic Learning in a Random World.* Springer. — Lei, J., et al. (2018). Distribution-free predictive inference for regression. *JASA* 113(523). — Romano, Y., Patterson, E., & Candès, E. (2019). Conformalized quantile regression. *NeurIPS*. — Angelopoulos, A., & Bates, S. (2023). Conformal prediction: a gentle introduction. *Foundations and Trends in Machine Learning* 16(4).

Learning objectives

By the end of this module you will be able to:

- **Explain** why classical prediction intervals need assumptions that conformal prediction drops.
- **State** exchangeability and **judge** when it is plausible.
- **Apply** split conformal prediction: compute $\hat{q}$ and the interval by hand and in Python.
- **Distinguish** marginal from conditional coverage — and say which one is guaranteed.
- **Build** adaptive intervals with conformalized quantile regression (CQR).
- **Construct** prediction sets for classifiers and read their sizes.
- **Avoid** the pitfalls: leakage into the calibration set, time series, tiny calibration sets.

Roadmap of this module

Uncertainty you can guarantee

1. Why a point prediction is a guess — and what Ch. 3's intervals assume.
2. Exchangeability: the one assumption conformal prediction needs.
3. Split conformal for regression: the whole recipe, and its proof sketch.
4. Marginal vs conditional coverage — the honest reading of the guarantee.
5. Adaptive intervals: locally weighted scores and CQR.
6. Prediction sets for classification; conformal vs the classical interval; pitfalls and production practice.

Outline

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A point prediction without uncertainty is a guess

Take the `Wage` data ($n = 3\,000$ workers): predict **wage** from **age**.

How much do 45-year-olds actually earn?

Among the 268 workers aged 44–46, wages average \$120 400 — but they range from \$36 700 to \$318 300, with a standard deviation of \$43 400. A model that answers "\$120k" communicates *none* of that spread: the single number is a guess dressed up as knowledge.

- Every model from this course — linear regression, boosting, a neural net — produces $\hat{f}(x)$. The question this module answers: *how far can the truth be from $\hat{f}(x)$, with a guarantee?*
- We want a recipe that works for **any** $\hat{f}$, on **finite** samples, **without** trusting the model.

Takeaway

A buyer sizing stock, a hospital planning beds, a grid operator buying reserve — none of them can act on "the forecast is 120". They act on "between 80 and 160, 9 times out of 10".

What Chapter 3 gave us — and what it silently assumed

The classical 90% prediction interval (Ch. 3)

Under the linear model $Y = X\beta + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, \sigma^2)$ i.i.d.: $\quad \hat{y}_0 \;\pm\; t_{n-p-1,\,0.95}\,\hat{\sigma}\sqrt{1 + x_0^\top (X^\top X)^{-1} x_0}.$

The fine print — three assumptions doing all the work

1. **Correct mean model**: $E[Y \mid X = x]$ really is linear in x.
2. **Gaussian errors**: the t/normal quantile calibrates the width.
3. **Homoskedasticity**: one σ for every x.

Break any of the three and the "90%" printed by `statsmodels` is a claim, not a fact.

Wage breaks the fine print

Wages are strongly right-skewed (skewness 1.68; 79 workers, 2.6%, earn above \$250k). The Gaussian recipe $\hat{f}(x) \pm 1.645\,\hat{\sigma}$ gives half-width 66.1 — and covers 95.2% of test wages while *claiming* 90%: miscalibrated and needlessly wide.

The conformal promise

What conformal prediction guarantees

From any fitted model $\hat{f}$ and a calibration set of n held-out points, build $C(x)$ such that for a new exchangeable point (X_{n+1}, Y_{n+1}):

$$\boxed{\Pr\left(Y_{n+1} \in C(X_{n+1})\right) \;\geq\; 1-\alpha}$$

Why this is remarkable — read the guarantee word by word

- **Distribution-free**: no Gaussian errors, no model for $P(Y \mid X)$ — wages can be as skewed as they like.
- **Finite-sample**: holds exactly at $n = 500$, not just as $n \to \infty$.
- **Any model**: $\hat{f}$ may be misspecified, overfit, or a black box.
- **Marginal**: averaged over the new point *and* the calibration draw — more on this honest word soon.

Takeaway

Conformal trades "if my model is right, coverage is exact at every x" for "*whatever* my model is, coverage is right *on average*".

Exercise A3.1 — What did we just buy? [Concept]

Exercise

1. Name the three assumptions behind the Chapter 3 prediction interval.
2. Which of the three does conformal prediction keep, and what does it demand instead?
3. A colleague says: "With conformal prediction, my 90% interval covers the true wage of *every 45-year-old* with probability 90%." Is that what the guarantee says?
4. Why does an overfit $\hat{f}$ not break the conformal guarantee (hint: where are the residuals computed)?

Solution — Exercise A3.1 (1/2)

Solution

Method. Compare the two guarantees assumption by assumption.

- **Part 1.** Correct (linear) mean model; Gaussian errors; homoskedastic errors (one σ for all x).
- **Part 2.** It keeps *none* of them. Instead it demands (i) **exchangeability** between calibration and test points and (ii) a **clean split**: $\hat{f}$ never sees the calibration data.

Solution — Exercise A3.1 (2/2)

Solution

- **Part 3.** No. The guarantee is **marginal**: averaged over all new points (and calibration draws), at least 90% are covered. Coverage *among 45-year-olds specifically* may be higher or lower — we will measure exactly this on Wage (98.6% for under-30s, 82.9% for 60-plus, same interval).
- **Part 4.** The scores are residuals on *held-out* calibration points. An overfit $\hat{f}$ produces *large* held-out residuals, so $\hat{q}$ grows and the interval honestly widens — the guarantee survives; only the width suffers.

Common mistake

Reading "90%" as a per-person (conditional) probability. Marginal coverage is a statement about the *average* over the population, not about your specific x.

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Exchangeability — the one assumption we need

Definition

Random variables $Z_1, \ldots, Z_m$ are **exchangeable** if for every permutation π,

$$(Z_{\pi(1)}, \ldots, Z_{\pi(m)}) \overset{d}{=} (Z_1, \ldots, Z_m).$$

What it means in plain words

- Every *ordering* of the data is equally likely — the labels "calibration point 17" and "test point" carry no information.
- **i.i.d.** $\Rightarrow$ **exchangeable**, but not conversely: exchangeability is *weaker* (e.g. sampling without replacement).
- Key consequence: among $n+1$ exchangeable scores with no ties, the **rank** of the last one is *uniform* on $\{1, \ldots, n+1\}$.

Takeaway

Conformal prediction is rank arithmetic on exchangeable scores. Everything follows from "the new point is nothing special".

The honest slide: what breaks exchangeability

- **Time series**: today's demand depends on yesterday's — orderings are *not* equally likely, and tomorrow may differ systematically.
- **Covariate shift**: calibrate on last year's customers, deploy on a new market — the X-distribution moved.
- **Feedback loops**: the model's own decisions (credit granted, price shown) change the data it will be judged on.

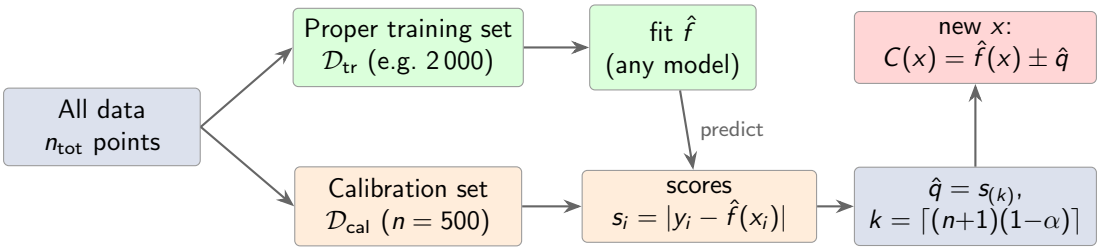
What still works

Violations degrade the guarantee *gradually*, and there are repairs: *weighted* conformal under known covariate shift (Tibshirani et al., 2019), blocked or online variants for dependence. This module sticks to the exchangeable case — the pitfalls section returns here.

Takeaway

Realised coverage is therefore also a *drift alarm*: sliding from 90% to 80% says the world has moved away from the calibration set, and it needs no labels beyond the ones you already collect.

Split conformal — the pipeline (native diagram)



Takeaway

One split, one model fit, one sorted list of residuals, one quantile. The model only ever sees $\mathcal{D}_{\text{tr}}$; the guarantee only ever uses $\mathcal{D}_{\text{cal}}$.

The whole recipe on one slide

Algorithm: split conformal prediction for regression

1. Split the data into $\mathcal{D}_{\text{tr}}$ and $\mathcal{D}_{\text{cal}}$ (sizes e.g. 2 000 and $n = 500$).
2. Fit $\hat{f}$ on $\mathcal{D}_{\text{tr}}$ *only*.
3. Compute calibration scores $s_i = |y_i - \hat{f}(x_i)|$ for $i \in \mathcal{D}_{\text{cal}}$.
4. Set $\boxed{\hat{q} = s_{(k)}, \qquad k = \lceil (n+1)(1-\alpha) \rceil}$ — the k-th smallest score.
5. For any new x report $C(x) = \big[\hat{f}(x) - \hat{q},\ \hat{f}(x) + \hat{q}\big]$.

Decoding the boxed formula

n = number of calibration points; α = tolerated miscoverage (0.1 for a 90% interval); $s_{(k)}$ = the k-th order statistic of the sorted scores; the ceiling $\lceil \cdot \rceil$ rounds *up*, so $\hat{q}$ is the $\frac{k}{n}$ empirical quantile — slightly *above* the naive $(1-\alpha)$-quantile. With $n = 500$, $\alpha = 0.1$: $k = \lceil 501 \times 0.9 \rceil = 451$, the 0.902 quantile.

Why the odd index? A tiny worked example

Nine calibration scores at 10% miscoverage

Scores: 1.2, 0.4, 2.8, 0.9, 1.7, 3.5, 0.6, 2.1, 1.0. Sorted: 0.4, 0.6, 0.9, 1.0, 1.2, 1.7, 2.1, 2.8, 3.5. Then $k = \lceil (9+1)(0.9) \rceil = 9 \Longrightarrow \hat{q} = s_{(9)} = 3.5$ (the maximum!). At $\alpha = 0.2$: $k = \lceil 10 \times 0.8 \rceil = 8$, so $\hat{q} = s_{(8)} = 2.8$. At $n = 19$, $\alpha = 0.1$: $k = \lceil 20 \times 0.9 \rceil = 18$ — the 18th of 19.

The intuition for the "$+1$"

The new point's score joins the n calibration scores as an $(n{+}1)$-th, equally-ranked member. To catch it with probability ≥ 0.9 we must clear at least 90% of *all* $n+1$ *ranks* — hence $(n+1)(1-\alpha)$, rounded up because ranks are integers. The naive 90% quantile of n scores would undercover, worst at small n.

Takeaway

Small calibration sets force conservative quantiles: with $n = 9$ the 90% interval must stretch to the *largest* observed residual.

The coverage guarantee, and why it is true

Theorem (split conformal coverage)

If the calibration pairs and the test pair are exchangeable, and $\hat{f}$ was fit without touching them, then

$$1-\alpha \;\le\; \Pr\left(Y_{n+1} \in C(X_{n+1})\right) \;\le\; 1-\alpha+\tfrac{1}{n+1},$$

the upper bound requiring almost-surely distinct scores.

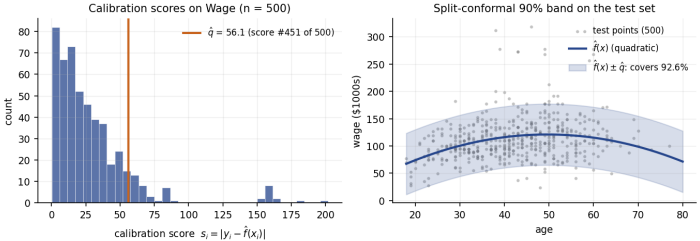
Proof sketch — three lines of rank arithmetic

1. Let $s_{n+1} = |Y_{n+1} - \hat{f}(X_{n+1})|$ join the n calibration scores. All $n+1$ scores are exchangeable (each is the same function of its own data point, with $\hat{f}$ fixed).
2. Coverage happens $\iff$ $s_{n+1} \le s_{(k)}$ $\iff$ $\mathrm{rank}(s_{n+1}) \le k$ among the $n+1$ scores.
3. That rank is uniform on $\{1,\dots,n+1\}$, so coverage $= \frac{k}{n+1} \ge \frac{(n+1)(1-\alpha)}{n+1} = 1-\alpha$. (Full argument: appendix.)

How tight is the guarantee at n of 500?

$k = 451 \Rightarrow$ coverage is pinned between 0.9 and $0.9 + \frac{1}{501} = 0.902$ — not just a bound, essentially an *equality*.

Wage: the calibration quantile and the band it buys



Takeaway

Quadratic fit on 2 000 training points; the 451st of 500 sorted calibration residuals gives $\hat{q} = 56.1$. The band $\hat{f}(\text{age}) \pm 56.1$ covers 92.6% of the 500 test wages — and is *narrower* than the miscalibrated Gaussian band (± 66.1, covering 95.2%).

Notebook §1: split conformal on Wage

`advanced_03_conformal_lab.ipynb`, §1 builds exactly this: the split (seed 2024), the sorted scores, $\hat{q} = 56.14$ and the test coverage 0.926 — about ten lines of numpy.

Exercise A3.2 — Computing the conformal quantile by hand [Math]

Exercise

A model is calibrated on $n = 9$ held-out points with absolute residuals

$$2.1, \quad 0.7, \quad 4.0, \quad 1.3, \quad 0.5, \quad 2.9, \quad 1.8, \quad 5.2, \quad 1.1.$$

1. Compute $\hat{q}$ for $\alpha = 0.2$ and write down $C(x)$ if $\hat{f}(x) = 100$.
2. Compute $\hat{q}$ for $\alpha = 0.1$. What do you notice?
3. For $n = 99$ and $\alpha = 0.05$: which order statistic is $\hat{q}$?
4. Exactly which coverage does the theorem guarantee in part 3 (both bounds)?

Solution — Exercise A3.2 (1/2)

Solution

Method. Sort, compute $k = \lceil (n+1)(1-\alpha) \rceil$, pick $s_{(k)}$. Sorted scores: 0.5, 0.7, 1.1, 1.3, 1.8, 2.1, 2.9, 4.0, 5.2.

- **Part 1.** $k = \lceil 10 \times 0.8 \rceil = 8 \Rightarrow \hat{q} = s_{(8)} = 4.0$, so $C(x) = [96.0,\ 104.0]$.
- **Part 2.** $k = \lceil 10 \times 0.9 \rceil = 9 \Rightarrow \hat{q} = s_{(9)} = 5.2$ — the *maximum*. With only 9 calibration points, a 90% guarantee forces the most conservative choice available.

Solution — Exercise A3.2 (2/2)

Solution

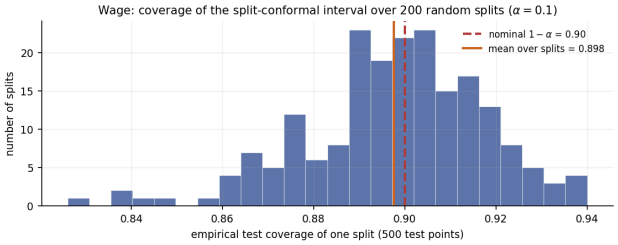
- **Part 3.** $k = \lceil 100 \times 0.95 \rceil = 95 \Rightarrow \hat{q}$ is the **95th smallest of 99** scores.
- **Part 4.** Coverage lies in $\left[\frac{95}{100},\ \frac{95}{100} + \frac{1}{100}\right] = [0.95,\ 0.96]$ — read off as $k/(n+1) = 95/100$ and $k/(n+1) + 1/(n+1)$.

Take-away. $\hat{q}$ is always a *single order statistic* — no interpolation, no model, just a sorted list and one index.

Common mistake

Computing the plain 90% empirical quantile ($0.9 \times 9 = 8.1$, so interpolating near $s_{(8)}$). The conformal index uses $n+1$, not n — forgetting the $+1$ is precisely what loses the finite-sample guarantee.

The guarantee is an average over splits — see it hold



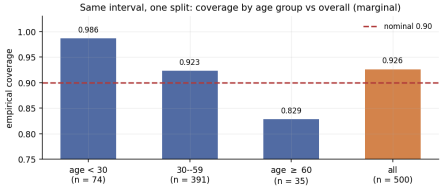
Takeaway

200 random splits of Wage: per-split test coverage fluctuates between 0.826 and 0.940 (sd 0.020), but the *mean* is 0.898 $\approx$ 0.9 — the marginal guarantee in action. Any *single* split can land at 0.87; nothing is wrong when it does.

Notebook §2: coverage over repeated splits

§2 re-runs the split 200 times and reproduces this histogram — mean 0.8976, sd 0.0198. The appendix derives why the spread is exactly of this size.

The honest slide: marginal is not conditional



Takeaway

Same interval, same split: coverage is 98.6% for under-30s but only 82.9% for the 60-plus group — the marginal 92.6% is an *average* that hides both. A constant-width band overcovers where wages are predictable and undercovers where they spread out.

Common mistake

Promising a customer segment "90% coverage" because the overall model is conformal. Per-subgroup coverage is **not** guaranteed — if a group matters, calibrate on that group (Mondrian conformal, appendix) or use adaptive scores (next section).

Exercise A3.3 — How small can the calibration set be? [Math]

Exercise

The conformal index is $k = \lceil (n+1)(1-\alpha) \rceil$, and $\hat{q} = s_{(k)}$ only exists if $k \leq n$.

1. Show that $k \leq n$ is equivalent to $n \geq \frac{1-\alpha}{\alpha} = \frac{1}{\alpha} - 1$.
2. Evaluate the minimal n for $\alpha = 0.1$, 0.05, and 0.01.
3. A team calibrates with $n = 10$ points at $\alpha = 0.05$. What interval does the recipe return, and why is that the honest answer?
4. Which coverage levels are *exactly* achievable with $n = 9$?

Solution — Exercise A3.3 (1/2)

Solution

Method. Unpack the ceiling: $k \le n \iff (n+1)(1-\alpha) \le n$.

- **Part 1.**
  $(n+1)(1-\alpha) \le n \iff n+1-\alpha n-\alpha \le n \iff 1-\alpha \le \alpha n \iff n \ge \frac{1-\alpha}{\alpha} = \frac{1}{\alpha} - 1.$
  ✓
- **Part 2.** $\alpha = 0.1$: $n \ge 9$. $\alpha = 0.05$: $n \ge 19$. $\alpha = 0.01$: $n \ge 99$.

Solution — Exercise A3.3 (2/2)

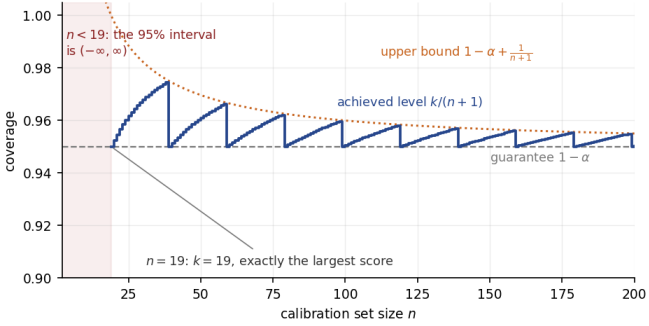
Solution

- **Part 3.** $k = \lceil 11 \times 0.95 \rceil = \lceil 10.45 \rceil = 11 > 10$: there is no 11th score, so $\hat{q} = \infty$ and $C(x) = (-\infty, \infty)$. Honest, because with 10 exchangeable scores the new point exceeds *all* of them with probability $1/11 \approx 9.1\% > 5\%$ — no finite cut-off can promise 95%.
- **Part 4.** Achievable exact levels are $\frac{k}{n+1} = \frac{k}{10}$: $10\%, 20\%, \ldots, 90\%, 100\%$ — coverage is *quantized* in steps of $\frac{1}{n+1}$.

Common mistake

Requesting $\alpha = 0.01$ from $n = 50$ calibration points and "fixing" the infinite interval by switching to the plain empirical maximum. The interval is then no longer guaranteed — collect more calibration data instead.

What the calibration set buys, drawn



Takeaway

Exercise A3.3 as a picture. Below $n = 19$ the 95% interval is $(-\infty, \infty)$; at exactly $n = 19$ the quantile is the largest score; and the overshoot above $1 - \alpha$ decays like $1/(n+1)$ — the sawtooth is the ceiling function doing its work.

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One width for everyone is a blunt instrument

The absolute-residual band on Wage is ± 56.1 *everywhere*: for 20-year-olds (tightly clustered wages) and for 60-year-olds (wide spread) alike.

Two ways to make the width follow the data

- **Locally weighted scores**: fit a second model $\hat{\sigma}(x)$ for the typical error size, and score

$$s_i = \frac{|y_i - \hat{f}(x_i)|}{\hat{\sigma}(x_i)} \qquad \Longrightarrow \qquad C(x) = \hat{f}(x) \pm \hat{q}\,\hat{\sigma}(x).$$

- **Conformalized quantile regression (CQR)**: model the *quantiles* of $Y \mid X$ directly, then calibrate them.

Both keep the marginal guarantee — any fixed score function works, because the scores of exchangeable points remain exchangeable.

Locally weighted conformal on Wage

$\hat{\sigma}(x)$: quadratic fit of |residual| on age. Result: $\hat{q} = 1.90$ (in σ-units), test coverage 89.6%, and widths now range from 42 (age 18) to 121 (age 50) instead of a flat 112.

The engine: quantile regression with the pinball loss

Pinball (quantile) loss at level τ

$$\rho_\tau(u) = \begin{cases} \tau\, u & u \geq 0 \\ (\tau - 1)\, u & u < 0 \end{cases} \qquad u = y - q.$$

Minimising $E\,\rho_\tau\big(Y - q(X)\big)$ over functions q yields the conditional τ-quantile of $Y \mid X$.

Why it works — in one sentence

Undershooting costs τ per unit, overshooting costs $1 - \tau$: the optimal $q(x)$ sits where a fraction τ of the mass lies below — exactly the quantile. At $\tau = 0.5$ it is the absolute loss, and we recover the median. Below, `alpha` *is* τ: the two fits trace the 5% and 95% edges of $Y \mid X$ — the raw band that CQR then calibrates.

```
from sklearn.ensemble import GradientBoostingRegressor # Ch. 8 engine
lo = GradientBoostingRegressor(loss="quantile", alpha=0.05, # 5% curve
       n_estimators=200, max_depth=2, learning_rate=0.05) # Ch. 8 knobs
hi = GradientBoostingRegressor(loss="quantile", alpha=0.95, # 95% curve
       n_estimators=200, max_depth=2, learning_rate=0.05) # same knobs
```

Conformalized quantile regression (CQR)

Algorithm (Romano, Patterson & Candès, 2019)

1. On $\mathcal{D}_{\text{tr}}$, fit quantile curves $\hat{q}_{\text{lo}}$ (level $\alpha/2$) and $\hat{q}_{\text{hi}}$ (level $1-\alpha/2$).
2. On $\mathcal{D}_{\text{cal}}$, score $\boxed{s_i = \max\left(\hat{q}_{\text{lo}}(x_i) - y_i,\ y_i - \hat{q}_{\text{hi}}(x_i)\right)}$
3. $\hat{q} = s_{(k)}$ with the usual $k = \lceil (n+1)(1-\alpha) \rceil$; report $C(x) = \left[\hat{q}_{\text{lo}}(x) - \hat{q},\ \hat{q}_{\text{hi}}(x) + \hat{q}\right]$.

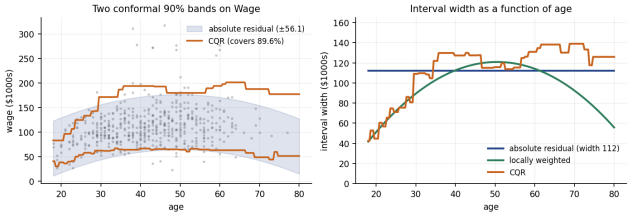
Decoding the score

s_i is the *signed distance to the nearer violated edge*: positive if y_i falls outside $[\hat{q}_{\text{lo}}, \hat{q}_{\text{hi}}]$ (by how much), *negative* if inside (how deep). If the fitted band already overcovers, $\hat{q} < 0$ *shrinks* it; if it undercovers, $\hat{q} > 0$ pads it. Same quantile index, same guarantee — the width now lives in the quantile curves.

Takeaway

CQR = "let Chapter 8 draw the band, let conformal calibrate it". On Wage the correction is tiny: $\hat{q} = 0.66$ — the boosted 5%/95% curves were almost calibrated already.

Wage: constant width vs adaptive width



Takeaway

Both bands cover $\approx$ 90% marginally (CQR: 89.6%), but CQR spends its width where the data spread: average width 70.5 for under-30s vs 134.9 for 60-plus (constant band: 112.3 for both). Note the locally weighted curve *falls* after age 60 — its quadratic $\hat{\sigma}$ extrapolates poorly: the allocation is only as good as the scale model, even though marginal coverage (89.6%) is untouched.

Notebook §3: adaptive intervals

§3 fits both adaptive variants and reproduces this figure's numbers: $\hat{q}_{\text{lw}} = 1.897$, $\hat{q}_{\text{cqr}} = 0.66$, and the three width profiles.

Extended Exercise A3.1 — CQR on Wage, end to end [Python]

Extended exercise (15 min)

Reproduce the adaptive band. Using the deck's split of `Wage` (seed 2024; 2 000 / 500 / 500):

1. Fit `GradientBoostingRegressor(loss="quantile")` at levels 0.05 and 0.95 on the training set (200 trees, depth 2, learning rate 0.05, `random_state=0`).
2. Compute the CQR scores on the calibration set and $\hat{q}$ at $\alpha = 0.1$; report test coverage.
3. Compare average interval width for ages < 30, 30–59 and ≥ 60 against the constant-width band (112.3).
4. Compare the two bands' *subgroup coverage* for under-30s and 60-plus. Which failure of the constant band does CQR repair, and what does it *not* guarantee?

Run this live

The worked solution sits in `advanced_03_conformal_lab.ipynb` under *Lecture exercises — worked Python solutions*: the cell headed *Extended Exercise A3.1 — CQR on Wage, end to end [Python]*.

Solution — Extended Exercise A3.1 (1/2)

Solution — code: fit / score / calibrate

```
import numpy as np, pandas as pd # data handling
from sklearn.ensemble import GradientBoostingRegressor # the CQR engine

wage = pd.read_csv("Wage.csv") # lab: via load()
x, y = wage["age"].to_numpy(float), wage["wage"].to_numpy() # x = age
rng = np.random.default_rng(2024) # the deck's split
i = rng.permutation(len(y)); tr, ca, te = i[:2000], i[2000:2500], i[2500:]
X = x.reshape(-1, 1) # sklearn wants 2-D
gb = dict(n_estimators=200, max_depth=2, learning_rate=0.05, random_state=0)
lo = GradientBoostingRegressor(loss="quantile", alpha=0.05, **gb).fit(X[tr], y[tr])
hi = GradientBoostingRegressor(loss="quantile", alpha=0.95, **gb).fit(X[tr], y[tr])
s = np.maximum(lo.predict(X[ca]) - y[ca], y[ca] - hi.predict(X[ca])) # CQR score
k = int(np.ceil((len(ca) + 1) * 0.9)) # k = 451 of 500
qhat = np.sort(s)[k - 1] # 0.66: tiny correction
L, U = lo.predict(X[te]) - qhat, hi.predict(X[te]) + qhat # pad both edges
print(round(qhat, 2), ((y[te] >= L) & (y[te] <= U)).mean()) # 0.66, cov 0.896
```

Solution — Extended Exercise A3.1 (2/2)

Solution — numbers and interpretation

Parts 2–3. $\hat{q} = 0.66$; test coverage $= 0.896$. Average widths:

	age < 30	age 30–59	age ≥ 60
Constant band (abs. residual)	112.3	112.3	112.3
CQR band	70.5	120.4	134.9

Part 4. Subgroup coverage on this split: constant band 0.986 (under-30) vs 0.829 (60-plus); CQR 0.797 vs 0.886. CQR repairs the *width misallocation* — it stops wasting 40 units of width on the young and reinvests them in the old, whose coverage rises from 0.829 to 0.886. **What it does not guarantee:** exact per-group coverage — the young bin ($n = 74$) lands at 0.797 here, within sampling noise of 0.9 but not pinned to it. Only the *marginal* 90% is guaranteed; per-group guarantees need per-group calibration.

Common mistake

Fitting $\hat{q}_{\text{lo}}, \hat{q}_{\text{hi}}$ on training *plus* calibration data "to use all the data". The scores are then not exchangeable with the test score, and the guarantee is void — the split is load-bearing.

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From intervals to prediction sets

For a classifier there is no interval — the honest object is a **set of classes** $C(x) \subseteq \{1, \dots, K\}$.

Split conformal for classification

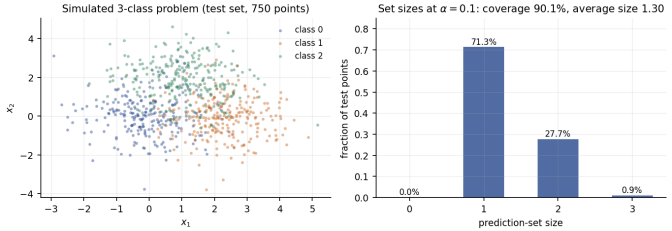
1. Fit a probabilistic classifier $\hat{p}$ on $\mathcal{D}_{\text{tr}}$ (e.g. softmax / logistic regression, Ch. 4).
2. Calibration scores: $\boxed{s_i = 1 - \hat{p}_{y_i}(x_i)}$ — one minus the probability assigned to the *true* class.
3. $\hat{q} = s_{(k)}$, $k = \lceil (n+1)(1-\alpha) \rceil$ as always; report

$$C(x) = \{\, k : \ \hat{p}_k(x) \geq 1 - \hat{q} \,\}.$$

Decoding the set rule

A class enters the set when its softmax probability clears the threshold $1 - \hat{q}$. Confident points get **singletons**; ambiguous points near a decision boundary get $\{1, 2\}$ or bigger. The guarantee: the *true* class is in $C(X)$ with probability $\geq 1 - \alpha$ — same rank argument, unchanged.

A 3-class simulation: sets earn their keep at the boundaries



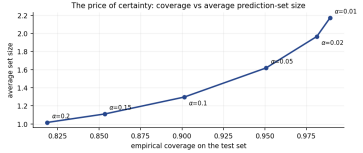
Takeaway

Three Gaussian classes (centres $(0,0)$, $(2.2,0)$, $(1.1,1.9)$; logistic accuracy 80.9%). At $\alpha = 0.1$: $\hat{q} = 0.7685$ (score 676 of 750), coverage 90.1%, average set size 1.30 — 71.3% singletons, 27.7% pairs, 0.9% full sets. The set *size* is a per-point difficulty meter the accuracy number hides.

Notebook §4: conformal classification

§4 simulates these three classes (seed 2024), builds the sets and reproduces coverage 0.9013 and the size distribution above.

The price of certainty: coverage vs set size



Takeaway

Demanding 99% coverage inflates the average set from 1.30 to 2.17 classes — two-thirds of the label space. Certainty is bought with set size; pick α by what a large set *costs downstream* (a manual review, a second lab test).

One slide on APS (adaptive prediction sets)

The $1 - \hat{p}$ score judges each class alone, so it can return *empty* sets when α is loose (here: 0.9% empty at $\alpha = 0.2$) and adapts poorly across easy/hard regions. **APS** scores the *cumulative* probability mass needed to reach the true class (add classes in decreasing $\hat{p}_k$); sets are never empty and track difficulty better, at the price of being larger on average. Same calibration machinery.

Exercise A3.4 — Reading prediction sets [Concept]

Exercise

A conformal classifier ($K = 3$, $\alpha = 0.1$) was calibrated with $\hat{q} = 0.7685$, so classes enter the set when $\hat{p}_k(x) \geq 0.2315$. Three test customers get softmax probabilities:

A: $(0.95,\ 0.03,\ 0.02)$ B: $(0.50,\ 0.45,\ 0.05)$ C: $(0.40,\ 0.35,\ 0.25)$

1. Write down the prediction set of each customer.
2. The plain classifier predicts class 0 for all three. What extra information do the sets carry?
3. Can the guarantee be read as "90% of *singleton* sets are correct"? Why not?
4. Under which circumstances can the rule return an *empty* set, and what should a production system do with one?

Solution — Exercise A3.4

Solution

Method. Threshold each probability at $1 - \hat{q} = 0.2315$.

- **Part 1.** A: $\{0\}$ (0.95 clears, others do not). B: $\{0, 1\}$ (0.50 and 0.45 clear). C: $\{0, 1, 2\}$ (all three ≥ 0.2315).
- **Part 2.** Identical point predictions, very different epistemic states: A is settled, B is a genuine two-way toss-up, C is anyone's guess. The set size is the honest "how sure are we" channel a point prediction lacks.
- **Part 3.** No. The guarantee is *marginal over all points*: $\geq 90\%$ of *all* sets contain the truth. Singletons are the *easy* points, so their hit rate is typically above 90%, while the large sets carry more of the mistakes — but neither is individually guaranteed.
- **Part 4.** Empty sets occur when even the largest probability misses the threshold: possible once $\hat{q} < 1 - \max_k \hat{p}_k(x)$, i.e. with loose α and diffident models (our simulation: 0.9% empty at $\alpha = 0.2$, none at $\alpha = 0.1$). Production systems treat empty (like oversized) sets as **abstentions**: route to a human.

Common mistake

Treating the set as a ranking. $C(x)$ is an *unordered* guarantee region — "class 0 first" is the old point prediction sneaking back in.

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Same data, two intervals: when the model is right, they agree

Homoskedastic simulation — the friendly case

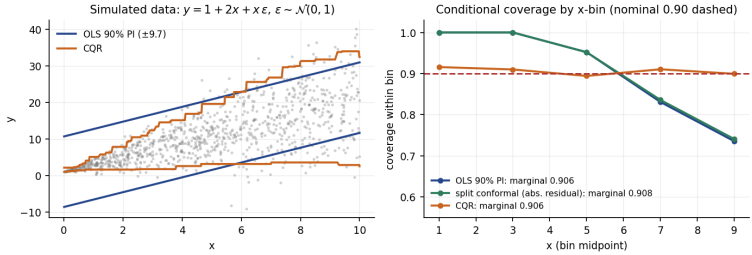
$y = 1 + 2x + \varepsilon$, $\varepsilon \sim \mathcal{N}(0, 2^2)$, $x \sim U(0, 10)$; 2 000 training, 1 000 calibration, 1 000 test points (seed 2024). Theory: 90% half-width $= 1.645 \times 2 = 3.29$.

	half-width	test coverage
OLS prediction interval (Ch. 3)	3.31	0.915
Split conformal (abs. residual)	3.24	0.909

Takeaway

When the Gaussian linear model is *true*, conformal costs essentially nothing: both intervals reproduce the theoretical ± 3.29. Conformal is not a better interval — it is the same interval with a *weaker warranty requirement*. That makes the pair a cheap diagnostic: run both, and if they disagree materially, the *model assumptions* are the prime suspect.

Heteroskedastic truth: where the 90% quietly stops being 90%



Takeaway

$y = 1 + 2x + x\,\varepsilon$: noise grows with x. All three methods hold $\approx 90\%$ *marginally* (OLS 0.906, conformal 0.908, CQR 0.906) — but conditionally the constant-width pair collapses to 73.5% (74.1%) for $x > 8$ while overcovering at 100% for $x < 2$. Only CQR stays near 0.90 in *every* bin (0.916 to 0.899). And only conformal *guarantees* its marginal number: over 100 repeated simulations, OLS averages 0.894 (luck, no warranty), conformal 0.900 (by theorem).

The failure that actually happens: plug-in intervals from a flexible model

Same heteroskedastic data — boosted regression

Fit a 500-tree gradient boosting model, then do what practitioners actually do: $\hat{f}(x) \pm 1.645 \times \hat{\sigma}_{\text{in}}$ with $\hat{\sigma}_{\text{in}}$ the sd of the *training* residuals.

	$\hat{\sigma}$ used	claimed	actual test coverage
Plug-in (training residuals)	3.92	90%	77.6%
Split conformal (same model)	$\hat{q} = 10.81$	$\geq 90\%$	90.7%

The truth: held-out residual sd is 6.12 — the model memorised a third of the noise, and the plug-in interval inherited the optimism.

Common mistake

Calibrating interval width on data the model has seen. Training residuals of a flexible model are systematically too small; every per-cent of memorised noise is a per-cent of silent undercoverage. The conformal split exists precisely to make this mistake impossible.

Chapter 3's interval vs split conformal — the comparison slide

	OLS prediction interval	**Split conformal**
Assumes	linear mean, Gaussian, homoskedastic errors	exchangeability + a clean split
Guarantee	exact at every x, *if* the model is true	marginal $\geq 1-\alpha$, finite-sample, *whatever* the model
Works with	the linear model it was derived for	any $\hat{f}$: boosting, nets, black boxes
Width	$\approx$ constant ($\hat{\sigma}$-driven; slightly wider at high leverage)	constant (abs. residual) or adaptive (CQR)
Cost	none	holds out n calibration points
Under misspecification	silently wrong coverage	coverage intact; only width reacts

Takeaway

Use the OLS interval when you *believe* the Gaussian linear model and want conditional statements. Use conformal when the model is flexible, the errors are ugly, or the coverage number will be *audited*.

Extended Exercise A3.2 — Stress-testing the two intervals [Integrative]

Extended exercise (15 min)

Simulate $y = 1 + 2x + x\,\varepsilon$, $\varepsilon \sim \mathcal{N}(0,1)$, $x \sim U(0,10)$, $n = 4\,000$ (split 2 000 / 1 000 / 1 000, seed 2024).

1. Fit OLS on the training set; build the 90% prediction interval $\hat{y} \pm 1.645\,\hat{\sigma}$ and measure test coverage overall, for $x < 2$, and for $x > 8$.
2. Build the split-conformal interval (absolute residuals, $\alpha = 0.1$) and measure the same three coverages. What does conformal fix here — and what does it not fix?
3. Fit a 500-tree `GradientBoostingRegressor`; report coverage of $\hat{f} \pm 1.645 \times \text{sd(training residuals)}$, then conformalise the same model. Explain the mechanism behind the gap.
4. Conclude: which of the three failure modes (skewed errors, heteroskedastic errors, optimistic $\hat{\sigma}$) does split conformal repair marginally, and which needs CQR?

Run this live

Worked solution in `advanced_03_conformal_lab.ipynb` under *Lecture exercises — worked Python solutions*, cell *Extended Exercise A3.2*.

Solution — Extended Exercise A3.2 (1/3)

Solution — code: the three intervals

```
import numpy as np, statsmodels.api as sm # OLS and arrays
from sklearn.ensemble import GradientBoostingRegressor # the flexible fit
rng = np.random.default_rng(2024) # the deck's seed
N = 4000; x = rng.uniform(0, 10, N); y = 1 + 2*x + x*rng.normal(0, 1, N)
tr, ca, te = np.arange(2000), np.arange(2000, 3000), np.arange(3000, 4000)
ols = sm.OLS(y[tr], sm.add_constant(x[tr])).fit() # 1. classical PI
half = 1.645 * np.sqrt(ols.scale) # pooled sigma: 5.87
pred = ols.params[0] + ols.params[1] * x[te] # test fitted values
cov = lambda hit: [hit.mean(), hit[x[te] < 2].mean(), hit[x[te] > 8].mean()]
print(cov(np.abs(y[te] - pred) <= half)) # [0.906, 1.000, 0.735]
k = int(np.ceil(1001 * 0.9)) # 2. conformal: k = 901
q = np.sort(np.abs(y[ca] - (ols.params[0] + ols.params[1]*x[ca])))[k-1]
print(q, cov(np.abs(y[te] - pred) <= q)) # 9.8 [0.908, 1.0, 0.741]
f = GradientBoostingRegressor(n_estimators=500, random_state=0) # 3. plug-in
f.fit(x[tr, None], y[tr]); sig = np.std(y[tr] - f.predict(x[tr, None])) # 3.92
print(cov(np.abs(y[te] - f.predict(x[te, None])) <= 1.645 * sig)) # [0.776, ...]
qg = np.sort(np.abs(y[ca] - f.predict(x[ca, None])))[k-1] # 10.81
print(cov(np.abs(y[te] - f.predict(x[te, None])) <= qg)) # [0.907, ...]
```

Output: cov returns [all, $x < 2$, $x > 8$] — marginal coverage first, then the two conditional ones.

Solution — Extended Exercise A3.2 (2/3)

Solution — interpretation

Part 1. OLS: marginal 0.906, but 100% for $x < 2$ and 73.5% for $x > 8$ — the pooled $\hat{\sigma} = 5.87$ is one compromise width for noise that ranges from ≈ 0 to 10. **Part 2.** Conformal ($\hat{q} = 9.79$): marginal 0.908, subgroups 1.00 / 0.741 — conformal *guarantees the marginal number* (over repeated data: 0.900 vs OLS's drifting 0.894) but a constant-width score cannot repair the *conditional* imbalance. That is CQR's job (it reaches 0.899 at $x > 8$).

Solution — Extended Exercise A3.2 (3/3)

Solution — interpretation

Part 3. Plug-in GBR: claimed 90%, actual 77.6% — training-residual sd (3.92) understates held-out error sd (6.12) because the 500-tree model partly memorised the training noise. Conformalising the *same* model restores 90.7% by pricing the width on unseen data ($\hat{q} = 10.81$). **Part 4.** Marginal repairs come free (skew, heteroskedasticity, optimism all handled — the guarantee never moved); *conditional* honesty across x is the one thing that needs an adaptive score (CQR / locally weighted).

Common mistake

Concluding from part 2 that conformal "failed". It delivered exactly what it promises — marginal coverage; expecting per-region coverage from a constant-width score is a specification error, not a method error.

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Pitfall 1: exchangeability violations in the wild

- **Time series**: calibration residuals from a five-year window, deployment two years later — trends, seasonality and regime shifts make the test point *special*, which is exactly what exchangeability forbids.
- **Covariate shift**: a marketing model calibrated on organic traffic, deployed during a paid campaign — the X-mix changed overnight.
- **Selection feedback**: the model approves loans, and only approved loans generate future calibration labels.

What to do instead

Time series: calibrate on the *most recent* block, recalibrate on a schedule, or use online/adaptive conformal that tunes α from realised coverage. Known covariate shift: *weighted* conformal with likelihood-ratio weights $w(x) = p_{\text{new}}(x)/p_{\text{cal}}(x)$. Always: monitor realised coverage — the guarantee you can check is the one you keep. A wind-power desk does exactly this, recalibrating on the trailing 30 days: a fixed calibration set from spring undercovers in autumn storms — same model, dead guarantee.

Pitfall 2: leakage between fitting and calibration

The guarantee has exactly one hard requirement inside your pipeline: *nothing about $\hat{f}$ may depend on the calibration set.*

The four common leak channels

1. **Preprocessing**: scaler / imputer / encoder fit on *all* data before the split.
2. **Tuning**: hyperparameters or early stopping selected on calibration performance.
3. **Feature selection**: screening variables on the full sample (the Ch. 5 classic).
4. **Duplicates**: the same customer's rows on both sides of the split.

Each one shrinks calibration scores below test scores $\Rightarrow$ $\hat{q}$ too small $\Rightarrow$ silent undercoverage — the plug-in GBR's 77.6% was this mechanism in its purest form.

Common mistake

"Refitting on train + calibration after computing $\hat{q}$, since the interval is already fixed." The deployed $\hat{f}$ is then not the one that generated the scores — the pair $(\hat{f}, \hat{q})$ is calibrated *together* or not at all.

From ten lines to production

Practical sizing rules

- **Feasibility**: α needs $n \geq \frac{1}{\alpha} - 1$ (19 points for 95%, 99 for 99%) — below that the honest interval is infinite (Exercise A3.3).
- **Stability**: with $n = 500$ the coverage of one split has sd $\approx 1.3\%$ (appendix); n between 500 and 1 000 is the usual sweet spot — beyond that, spend the data on training.
- **Report the nature of the guarantee**: marginal, at the stated α, under exchangeability — three words that pre-empt three misreadings.

The production tool

Teams reach for **MAPIE** (*Model Agnostic Prediction Interval Estimator*, scikit-learn compatible): split/cross conformal, CQR and classification sets behind one API, plus jackknife+ variants that recycle training folds. It is not installed here — deliberately: everything in this module was ten lines from scratch, and those ten lines are the audit trail you will be asked for.

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Python lab — run it live in the notebook

Companion notebook: `advanced_03_conformal_lab.ipynb`

The code in this section is meant to be **demonstrated live**. Open the companion Jupyter notebook and run it cell by cell:

- §1, *Split conformal on Wage* — the ten-line recipe: $\hat{q} = 56.14$, coverage 0.926;
- §2, *Coverage over repeated splits* — 200 splits, mean 0.898;
- §3, *Adaptive intervals* — locally weighted scores and CQR, width profiles by age;
- §4, *Conformal classification* — 3-class simulation, prediction sets, size distribution;
- §5, *OLS prediction interval vs conformal* — the friendly homoskedastic case (the heteroskedastic stress test is worked as Extended Exercise A3.2).

Data loads via the `ISLP` package or the bundled CSV files; §6, *Exercises* ends the notebook with hands-on tasks.

Split conformal in ten lines

```
import numpy as np, pandas as pd # arrays + CSV
from sklearn.linear_model import LinearRegression # any model works

wage = pd.read_csv("Wage.csv") # lab: via load()
x, y = wage["age"].to_numpy(float), wage["wage"].to_numpy() # x = age
rng = np.random.default_rng(2024) # reproducible split
i = rng.permutation(len(y)); tr, ca, te = i[:2000], i[2000:2500], i[2500:]
P = lambda a: np.column_stack([a, a**2]) # quadratic in age
f = LinearRegression().fit(P(x[tr]), y[tr]) # 1. fit on TRAIN only
s = np.abs(y[ca] - f.predict(P(x[ca]))) # 2. calibration scores
k = int(np.ceil((len(ca) + 1) * (1 - 0.1))) # 3. k = ceil((n+1)(1-a))
qhat = np.sort(s)[k - 1] # k-th smallest score
cov = (np.abs(y[te] - f.predict(P(x[te]))) <= qhat).mean() # 4. coverage
print(k, qhat.round(2), cov) # 451 56.14 0.926
```

How to read this

Key call: `np.sort(s)[k-1]`, the 451st smallest held-out residual: $\hat{q} = 56.14$, so $\hat{f}(x) \pm 56.14$ covers 92.6%.

Exercise A3.5 — Conformal classification from scratch [Python]

Exercise

Build prediction sets for the deck's 3-class simulation (three Gaussian classes, 1 000 points each, centres $(0,0)$, $(2.2,0)$, $(1.1,1.9)$, unit variance, seed 2024; split 1 500 / 750 / 750).

1. Fit `LogisticRegression` on the training set.
2. Compute the calibration scores $s_i = 1 - \hat{p}_{y_i}(x_i)$ and $\hat{q}$ at $\alpha = 0.1$ (which k?).
3. Build the test-set prediction sets and report coverage, average set size, and the share of singleton sets.
4. Re-run with $\alpha = 0.01$: what happens to the average set size?

Run this live

Worked solution in `advanced_03_conformal_lab.ipynb` under *Lecture exercises — worked Python solutions*, cell *Exercise A3.5*.

Solution — Exercise A3.5

Solution — code and expected numbers

```
import numpy as np # arrays only
from sklearn.linear_model import LogisticRegression # Ch. 4 model
rng = np.random.default_rng(2024) # deck's seed
mu = np.array([[0, 0], [2.2, 0], [1.1, 1.9]]) # class centres
X = np.vstack([rng.normal(m, 1.0, (1000, 2)) for m in mu]) # 1000 each
y = np.repeat([0, 1, 2], 1000) # labels, in order
p = rng.permutation(3000); X, y = X[p], y[p] # shuffle once
clf = LogisticRegression(max_iter=1000).fit(X[:1500], y[:1500]) # TRAIN
pc = clf.predict_proba(X[1500:2250]) # calibration probs
s = 1 - pc[np.arange(750), y[1500:2250]] # score: 1 - p_true
k = int(np.ceil(751 * 0.9)) # k = 676 of 750
qhat = np.sort(s)[k - 1] # qhat = 0.7685
sets = clf.predict_proba(X[2250:]) >= 1 - qhat # keep p >= 0.2315
cov = sets[np.arange(750), y[2250:]].mean() # truth in the set?
print(qhat.round(4), cov, sets.sum(1).mean().round(3)) # 0.7685 0.9013 1.296
```

Parts 3–4. Coverage 0.9013, average size 1.30, singletons 71.3%. At $\alpha = 0.01$: $k = \lceil 751 \times 0.99 \rceil = 744$, $\hat{q} = 0.9638$, coverage 98.9% — and the average set grows to 2.17 of 3 classes.

Common mistake

Scoring with $1 - \max_k \hat{p}_k$ (the *predicted* class) instead of $1 - \hat{p}_{y_i}$ (the *true* class). The score must know the truth on the calibration set — that is what it is for.

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Module A3 in one slide

What we accomplished

Turned any point predictor from this course into a set-valued predictor with a finite-sample coverage guarantee.

- Exchangeability — the single assumption, and what breaks it.
- Split conformal: scores, $k = \lceil (n+1)(1-\alpha) \rceil$, interval $\hat{f}(x) \pm \hat{q}$; proof by rank arithmetic.
- The guarantee is *marginal* — averaged over points and splits, not per subgroup.
- Adaptive widths: locally weighted scores and conformalized quantile regression.
- Classification: prediction sets, set size as a difficulty meter, APS.
- Versus Chapter 3: agreement under the Gaussian model, guarantees under misspecification, and the plug-in trap.

Key formulas at a glance

Conformal index $k = \lceil (n+1)(1-\alpha) \rceil$; $\quad \hat{q} = s_{(k)}$; $\quad$ finite iff $n \geq \frac{1}{\alpha} - 1$.

Regression interval $s_i = |y_i - \hat{f}(x_i)|$; $\quad C(x) = [\hat{f}(x) - \hat{q},\ \hat{f}(x) + \hat{q}]$.

Coverage guarantee $1 - \alpha \ \leq \ \Pr(Y_{n+1} \in C(X_{n+1})) \ \leq \ 1 - \alpha + \frac{1}{n+1}$ (distinct scores).

Locally weighted $s_i = |y_i - \hat{f}(x_i)| / \hat{\sigma}(x_i)$; $\quad C(x) = \hat{f}(x) \pm \hat{q}\,\hat{\sigma}(x)$.

CQR $s_i = \max(\hat{q}_{\text{lo}}(x_i) - y_i,\ y_i - \hat{q}_{\text{hi}}(x_i))$; $\quad C(x) = [\hat{q}_{\text{lo}}(x) - \hat{q},\ \hat{q}_{\text{hi}}(x) + \hat{q}]$.

Classification set $s_i = 1 - \hat{p}_{y_i}(x_i)$; $\quad C(x) = \{k : \hat{p}_k(x) \geq 1 - \hat{q}\}$.

Pinball loss $\rho_\tau(u) = \tau u\,[u \geq 0] + (\tau - 1)u\,[u < 0]$ — minimised by the τ-quantile.

Vocabulary checklist

Term	Meaning
Exchangeability	every ordering of the data equally likely (weaker than i.i.d.)
Calibration set	held-out data that prices the interval width
Conformity score	how badly a point disagrees with the model's prediction
$\hat{q}$	the $\lceil (n+1)(1-\alpha) \rceil$-th smallest calibration score
Marginal coverage	$\geq 1-\alpha$ on average over points and splits
Conditional coverage	coverage at a specific x / subgroup — *not* guaranteed
Locally weighted CP	residuals scaled by a fitted error-size model
CQR	conformalized quantile regression — calibrated quantile band
Prediction set	set of classes guaranteed to contain the truth
APS	adaptive prediction sets via cumulative probability mass
Weighted / Mondrian CP	repairs for covariate shift / per-group guarantees

Decision rules of thumb

- Any black-box model, honest uncertainty needed $\Rightarrow$ **split conformal** — ten lines, one held-out slice.
- Spread visibly varies with $x \Rightarrow$ **CQR** (or locally weighted scores); constant width wastes coverage on the easy region.
- Classifier feeding a triage / review queue $\Rightarrow$ **prediction sets**; route big or empty sets to a human.
- Believe the Gaussian linear model and need per-x statements $\Rightarrow$ Chapter 3's interval is fine — and conformal is a free cross-check.
- Time series or known shift $\Rightarrow$ recent-block calibration, weighted or online conformal; *never* vanilla split conformal untouched.
- Sizing: $n_{\text{cal}} \approx 500$–$1\,000$; feasibility floor $n \geq 1/\alpha - 1$.
- A subgroup matters on its own $\Rightarrow$ calibrate within the subgroup.

Common pitfalls

Don't

1. Tune hyperparameters, early stopping, scalers or features on the calibration set — that is leakage, and it undercovers silently.
2. Read "90% marginal" as "90% for every customer segment".
3. Use training residuals as calibration scores (the 77.6% trap).
4. Apply vanilla split conformal to time series and call it guaranteed.
5. Demand $\alpha = 0.01$ from 50 calibration points — infeasible by arithmetic.
6. Keep last year's $\hat{q}$ after retraining the model or after the world drifts — the pair $(\hat{f}, \hat{q})$ is calibrated together.

Self-check questions

Can you answer these without looking back?

1. Why $\lceil (n+1)(1-\alpha) \rceil$ and not the plain $(1-\alpha)$-quantile of the n scores?
2. State the two-sided coverage bound. Where does each side come from?
3. Your single split shows coverage 0.87 at $\alpha = 0.1$. Is the method broken?
4. What exactly does CQR's $\max(\hat{q}_{\text{lo}} - y,\ y - \hat{q}_{\text{hi}})$ score measure, and when is $\hat{q}$ negative?
5. Why can a prediction set be empty under the $1 - \hat{p}$ score, and which score family avoids that?
6. Name the four leakage channels between fitting and calibration.

Where to look

1–2: Split Conformal section (and the appendix proof). 3: the coverage histogram. 4: Adaptive Intervals. 5: the APS slide. 6: Pitfall 2.

Five things to remember

Module A3 in five lines

1. **One assumption: exchangeability.** No Gaussianity, no correct model — but time order, distribution shift and feedback all break it.
2. **Split conformal is rank arithmetic:** the $k = \lceil (n+1)(1-\alpha) \rceil$-th calibration score gives $\hat{f}(x) \pm \hat{q}$ with finite-sample coverage.
3. **The guarantee is marginal** — averaged over points and splits, not per subgroup and not conditional on x.
4. **Honest width is not informative width** — locally weighted scores and CQR make the interval track difficulty.
5. **Classification gives prediction sets**, and the set size is the difficulty meter.

What's next

- **Module A1 — Randomised Controlled Trials**: marginal coverage is a guarantee about *predictions*, not interventions; A1 builds the design that answers "what happens if *we* act?" (read module A7 first — A1 leans on it).
- **Module A2 — Explainable AI with Shapley Values**: the interval says how much to trust a prediction; A2 says which features built it.

Appendix — what is in here, and why

Nothing in the appendix is needed to follow the module: each slide is a formal derivation, a longer worked exercise, or a side topic. Read it when you want the full story, or when a later chapter sends you back here.

Contents of the appendix

- **The coverage theorem, proved in full** — the rank argument with both bounds and the tie-breaking detail; the main deck only sketched it.
- **How coverage fluctuates: the Beta law** — the exact distribution of one split's coverage, moved out because it needs the Beta distribution.
- **Full conformal and jackknife+** — what you gain and pay when you refuse to split; a side topic beyond the split recipe.
- **Mondrian (group-conditional) conformal** — the fix for the subgroup problem; deferred until the marginal story was complete.
- **A drill exercise on the Beta law** — connects the histogram of the main deck to the formula here.

The coverage theorem, proved in full

Setup. Scores $s_1, \ldots, s_n$ (calibration) and s_{n+1} (test) are exchangeable; $k = \lceil (n+1)(1-\alpha) \rceil$; coverage event $A = \{s_{n+1} \le s_{(k)}\}$ with $s_{(k)}$ the k-th smallest of $s_1, \ldots, s_n$.

Step 1 — the event is a rank statement

Assume first no ties. Let r be the rank of s_{n+1} among all $n+1$ scores. Exactly $r-1$ calibration scores lie below s_{n+1}, hence $s_{n+1} \le s_{(k)} \iff r - 1 \le k - 1 \iff r \le k$.

Step 2 — the rank is uniform

Exchangeability makes every assignment of the $n+1$ scores to the $n+1$ ranks equally likely, so $\Pr(r = j) = \frac{1}{n+1}$ for each j and $\Pr(A) = \frac{k}{n+1}$.

Step 3 — both bounds

Lower: $k \ge (n+1)(1-\alpha) \Rightarrow \Pr(A) \ge 1 - \alpha$. Upper: $k < (n+1)(1-\alpha) + 1 \Rightarrow \Pr(A) < 1 - \alpha + \frac{1}{n+1}$. **Ties:** with ties, break them by adding i.i.d. uniform jitter to every score — exchangeability is preserved, the jittered interval is (weakly) inside the original one, so the *lower* bound survives; only the upper bound needs the distinct-scores assumption.

How coverage fluctuates from split to split: the Beta law

Conditional coverage of one calibration set

Given the calibration set, the interval's true coverage is $F(\hat{q})$, where F is the score distribution. For continuous F, $F(s_{(k)}) \sim \text{Beta}(k,\ n+1-k)$, hence

$$E\big[F(\hat{q})\big] = \frac{k}{n+1}, \qquad \text{sd} = \sqrt{\frac{k\,(n+1-k)}{(n+1)^2(n+2)}}.$$

Predicting the Wage histogram before running it

$n = 500$, $k = 451$: mean $= \frac{451}{501} = 0.9002$, sd $= 0.0134$. One split's *measured* coverage adds test-set noise ($\sqrt{0.9 \times 0.1/500} = 0.0134$); combined: $\sqrt{0.0134^2 + 0.0134^2} = 0.019$. Observed over 200 splits: sd $= 0.0198$. Theory and histogram agree to the third decimal.

Takeaway

The guarantee fixes the *mean* of the coverage distribution; the Beta law fixes its *spread*. Doubling n shrinks the sd by $\sqrt{2}$ — this is the calculation behind the "$n \approx 500$–$1\,000$ suffices" rule of thumb.

Refusing to split: full conformal and jackknife+

Method	Idea	Model fits	Guarantee
Split conformal	one split, one fit	1	$\geq 1-\alpha$
Full conformal	for every candidate y, refit including (x, y) and test its score's rank	one per candidate y (a grid)	$\geq 1-\alpha$
Jackknife+	n leave-one-out fits; intervals from LOO residuals	n	$\geq 1-2\alpha$
CV+	K-fold version of jackknife+	K	$\geq 1-2\alpha$

Why this module taught the split version

Full conformal uses every point for both fitting and calibration — statistically the most efficient, computationally absurd for boosting or neural nets. Jackknife+/CV+ sit in between, with a weaker worst-case constant ($1-2\alpha$; in practice close to $1-\alpha$). Split conformal costs one held-out slice and *one* model fit — which is why it is the production default and the MAPIE default.

Mondrian conformal: buying back the subgroups

Group-conditional (Mondrian) conformal

Partition the space into groups $g(x)$ fixed *in advance* (age bands, product lines, hospitals). Calibrate a separate $\hat{q}_g$ from the calibration scores of each group. Then, for every group, $\Pr\left(Y \in C(X) \mid g(X) = g\right) \geq 1 - \alpha$.

The price

Each group needs its own feasible calibration count ($n_g \geq \frac{1}{\alpha} - 1$, and comfortably more for stable widths — the Beta law now applies *per group*). On Wage, the 60-plus group had only 35 test points and ~ 50 calibration points per split at $\alpha = 0.1$: feasible, but with visibly noisy widths. Fully *conditional* coverage at every single x is provably impossible distribution-free — groups are the honest middle ground.

Takeaway

Marginal coverage is free; group coverage costs calibration data per group; pointwise coverage cannot be bought at any price (without assumptions).

Appendix Exercise A3.A — The Beta law in your hands [Math]

Exercise

A team calibrates with $n = 99$ points at $\alpha = 0.1$.

1. Compute k and the exact expected coverage $k/(n+1)$.
2. Using the Beta law, compute the sd of the true coverage of one calibration draw. (Hint: $\text{sd} = \sqrt{k(n+1-k)}/\big((n+1)\sqrt{n+2}\big)$.)
3. The team wants the sd below 0.01. Roughly which n do they need (use $\text{sd} \approx \sqrt{\alpha(1-\alpha)/n}$)?

Solution — Appendix Exercise A3.A

Solution

Method. Plug into the Beta$(k,\ n+1-k)$ moments.

- **Part 1.** $k = \lceil 100 \times 0.9 \rceil = 90$; expected coverage $= \frac{90}{100} = 0.900$ exactly.
- **Part 2.** $\text{sd} = \sqrt{90 \times 10}\,/\,\big(100\sqrt{101}\big) = 30/1004.99 \approx 0.030$: a single calibration draw's true coverage typically sits anywhere in ≈ 0.87–0.93.
- **Part 3.** $\sqrt{0.09/n} \le 0.01 \iff n \ge 900$. Nine hundred calibration points buy percent-level stability — consistent with the deck's 500–1 000 rule of thumb ($n = 500$ gave sd 0.0134).

Common mistake: treating the guarantee's *mean* statement as if it removed this spread. The theorem constrains the average over calibration draws; the Beta law says how far your *one* draw can sit from it.